

Impervious surface density from Sentinel-2 composites and AlphaEarth embeddings: Milan, Hanoi and Ho Chi Minh City, 2018

1 Introduction

Impervious surface density is the share of a pixel that is sealed against infiltration, and it is the quantity this report maps for three cities in 2018: Milan, Hanoi and Ho Chi Minh City. The work supports the ITALY-VIETNAM project on Local Climate Zones, Urban Heat Island and Geomatics (LCZ-UHI-GEO, CUP D47G24000110001), which is why those three cities and that year. A high-resolution imperviousness layer is an input to that project's wider objectives, being relevant to local climate zone characterisation and to urban heat island analysis in the Vietnamese cities [1, p. 11].

The study extends an earlier one. Mapping impervious density from AlphaEarth satellite embeddings in Milan was established by Žgela, whose report is the companion to this one and is assumed read. This project changes three things. It adds explicit Sentinel-2 composites as an alternative predictor set, so that a learned general-purpose representation can be compared against reflectance statistics chosen for this task. It extends the work to two Vietnamese cities, which turns a single-city study into a test of whether a model transfers. And it adds a second validation against photo-interpreted plots that no model saw, which is what allows the difference between agreement and accuracy to be measured rather than assumed.

Attribution

The AlphaEarth method and the shared Milan sample set are Žgela's work and are used here as supplied. The 3 500-point stratified sample, the spatial blocking and buffering that separate training from test, and the treatment of the 64-dimensional embedding are all his design, described in Sections 2.2 and 3.1 with citations rather than re-derived. Several results in this report are reproductions of his and are labelled as such where they appear: the cross-validation against holdout reversal in Section 4.2, the per-class pattern of local retraining in Section 5.2, and the qualitative observation behind the bias recovery of Section 7.3.

The embeddings pipeline was independently re-run in this project. It reproduces his published holdout figures exactly, for both estimators. The random forest gives RMSE 14.123, MAE 10.675, R^2 0.837 and bias +0.624 against his reported 14.12, 10.68, 0.837 and +0.62; the support vector regressor gives 14.756, 11.400, 0.822 and +0.274 against his 14.76, 11.40, 0.822 and +0.27 [1, p. 5]. Every figure matches to the precision at which he published it.

This is recorded as verification that the shared baseline is correctly reconstructed, and it is not a new result. Its purpose is narrow and worth stating plainly: every comparison in Sections 4 and 7 measures a new predictor set against that baseline, so a baseline that had drifted would move those comparisons without announcing itself. The reproduction establishes that it has not drifted. Nothing about the embeddings is being claimed here beyond what Žgela already established.

What this report finds

The results are summarised here because two of them qualify each other, and a reader who takes the first without the second will overstate what the study shows.

Against CLMS in Milan, all three Sentinel-2 composites outperform the AlphaEarth embeddings, and the margin is wide. The percentile composite reaches an RMSE of 9.462 against the embeddings' 14.123, a reduction of 33 %, with the same ordering on RMSE, MAE and R^2 . The advantage is also mechanistically interpretable: it comes from the low percentiles of the red band and the high percentiles of the near infrared, which are precisely the quantiles a median composite discards.

That margin is measured against the product the models were trained on, and it does not survive the change of reference intact. Scored against photo-interpreted plots instead, the four-way spread between predictor sets compresses about 6.4-fold, from 33.0 % to 5.2 % of the worst map, and the ordering partly reshuffles. Most of what the first validation measured as separation between predictor sets was agreement with CLMS rather than accuracy. The compression is not a dissolution, however: paired testing on the same plots still resolves the four maps into two distinguishable tiers. The honest statement, developed in Section 7.1, is that the advantage shrinks sharply and remains real.

Zero-shot transfer of a Milan-trained model to either Vietnamese city fails. The transferred models score R^2 at or below zero in three of the four city and predictor combinations, which is no more informative than predicting the mean everywhere, and local retraining recovers R^2 to between 0.522 and 0.649 in every case. The failure has a distributional mechanism rather than merely a calibration one: the transferred embedding maps lose the low end of the predicted distribution almost entirely, retaining 3.1 % of the reference's sub-20 % mass in Hanoi and none of it in Ho Chi Minh City, and every local retrain restores it. Retraining is not uniformly better, though. Above 80 % imperviousness it is worse than the transferred model, which Section 5.2 reports rather than sets aside.

Both training products, scored against photo-interpretation on the same terms as the models, do no better than the models fitted to them. That measures the quality of the labels and not a ceiling on the maps, and the two are distinct: a model can outperform its own training target, and here it measurably does. GHS-BUILT-S under-marks the interpreted reference by close to 20 percentage points in both Vietnamese cities, largely because it excludes roads by design, while the models fitted to it recover between 38 % and 67 % of that deficit.

How to read this report

Two conventions carry through and are set out where they are defined, in Sections 3.3 and 3.4.

The first is that every map is scored twice. Same-source validation scores a map against the product it was trained on and measures agreement with that target. Independent validation scores it against the photo-interpreted plots and measures accuracy against a reference no model saw. Both are necessary, because the first cannot distinguish a good map from one that merely resembles its labels. They are never merged, and their numbers are never compared directly: they are scored against different references on different samples, so a difference between one and the other would measure the change of reference rather than anything about a map. Sections 4 and 5 report the first, Section 6 the second, and Section 7.1 is the one place they are set side by side, with the relationship between them as its subject.

The second is the sign of the bias. Bias is observed minus predicted throughout, so a positive value means the reference is higher than the map and the map under-predicts. The convention is identical in both validations, and the two produce biases of opposite sign for reasons that are not a contradiction: Section 5 reports negative biases where models read higher than GHS-BUILT-S, and Section 6 reports positive ones where GHS-BUILT-S reads lower than the interpreted reference. Those are different comparisons and the signs must not be read across them.

Section 8 sets out five constraints that bound what the results support, each stated with the conclusion it forbids. Two are worth flagging in advance because they limit comparisons a reader might otherwise expect this report to make. The Vietnamese Sentinel-2 composites are built from far fewer usable dates than Milan’s, so composite depth is confounded with predictor type there and the ordering between the two predictor sets in Vietnam is not a controlled comparison. And the validation plots are stratified rather than drawn at random over each city, so the independent metrics compare maps against one another but do not estimate city-wide accuracy.

2 Data

Four data sets enter this study: two impervious-density products used as training targets, two predictor sets, and one set of photo-interpreted plots used for validation alone. The first three are treated as Žgela treats them and are described here only as far as is needed to follow the results; the fourth is new to this project and is described in full.

2.1 Targets: CLMS and GHS-BUILT-S

The models are fitted to two different products. Milan uses the Copernicus Land Monitoring Service imperviousness density layer for 2018 [2], and Hanoi and Ho Chi Minh City use GHS-BUILT-S for the same year [3]. Both are supplied at 10 m and both report a percentage per pixel, so they enter the modelling identically. What they measure is not the same thing, and the difference is not noise.

CLMS measures sealed surface. Its target is the fraction of a pixel that water cannot pass through, which includes roads, pavements, car parks and rooftops alike. GHS-BUILT-S measures built-up surface, which is a narrower quantity. Žgela states the consequence twice in his report and it is taken from him rather than asserted independently here: the product does not include all impervious surfaces such as roads, accounting only for built-up surfaces, namely buildings and a limited set of other roofed infrastructure, and he describes this as an inherent limitation of the chosen reference dataset [1, pp. 8, 10].

Roads are therefore in the Milan target and largely absent from the Vietnamese one. This matters later. Section 6 scores both products against photo-interpretation on the same terms as the models, and Section 7.3 measures what the models recover of the resulting deficit. A reader who meets the GHS-BUILT-S result there without having met this paragraph will read a definitional mismatch as a measurement error.

The choice of GHS-BUILT-S was nonetheless the right one, and was made for a stated reason rather than by default. Žgela records that no other suitable 2018 reference product existed for the two Vietnamese cities. Matching the label year to the imagery year was treated as the binding constraint, and a target with a known definitional gap was preferred to a target from the wrong year or to no Vietnamese experiment at all.

2.2 AlphaEarth embeddings

The first predictor set is the AlphaEarth satellite embedding [4], GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL, a 64-dimensional learned representation supplied at 10 m. The collection is filtered to a single year and all 64 bands, A00 to A63, are taken as predictors. The method and its rationale are Žgela’s [1, pp. 3–5] and are not re-derived here.

Two properties of that treatment matter for what follows.

The first is that all 64 bands are used, without feature selection. Žgela tested every band against the IMD groups by Kruskal-Wallis [5] and found 60 of the 64 to differ significantly across classes, and a separate feature-selection experiment found the full set to outperform any reduced subset [1, pp. 3–5, Fig. 3].

The embeddings are therefore used whole in this project rather than pruned, and the obvious question of whether a smaller subset would do as well has already been answered in the negative on this same data.

The second is coverage. The embedding is an annual product and the same one-year filter is applied in Milan, Hanoi and Ho Chi Minh City, so all three cities receive identical treatment with no seasonal gap. Section 2.3 shows that the Sentinel-2 side has no such symmetry, and Sections 5 and 7 both depend on the difference: the embeddings transfer experiment compares annual coverage against annual coverage, and the Sentinel-2 one does not.

2.3 Sentinel-2: search, screening and composites

The second predictor set is built from Sentinel-2 surface reflectance [6], COPENICUS/S2_SR_HARMONIZED, over each city's area of interest for 2018. Ten bands are carried: B2, B3, B4, B5, B6, B7, B8, B8A, B11 and B12. The four visible and near-infrared bands are native 10 m and the remainder are resampled to the same grid.

Screening is per acquisition date and applies three criteria, each computed over the city's area of interest after a scene classification mask that removes cloud, cloud shadow, cirrus, snow, saturated and no-data pixels, with a cloud probability gate at $MSK_CLDPRB < 40$ on top. A date is kept when cloud cover over the area of interest is below a threshold, when the share of valid pixels is above a threshold, and when scene coverage of the area of interest is at least 95 %. Milan passes at the strict setting: cloud below 20 % and valid above 90 %. Neither Vietnamese city yields a workable set at that setting, so both are screened one rung looser, at cloud below 35 % and valid above 80 %.

What survives is very unequal. Milan retains 30 usable dates. Hanoi retains 4 and Ho Chi Minh City 3. Figure 1 shows them on the 2018 calendar.

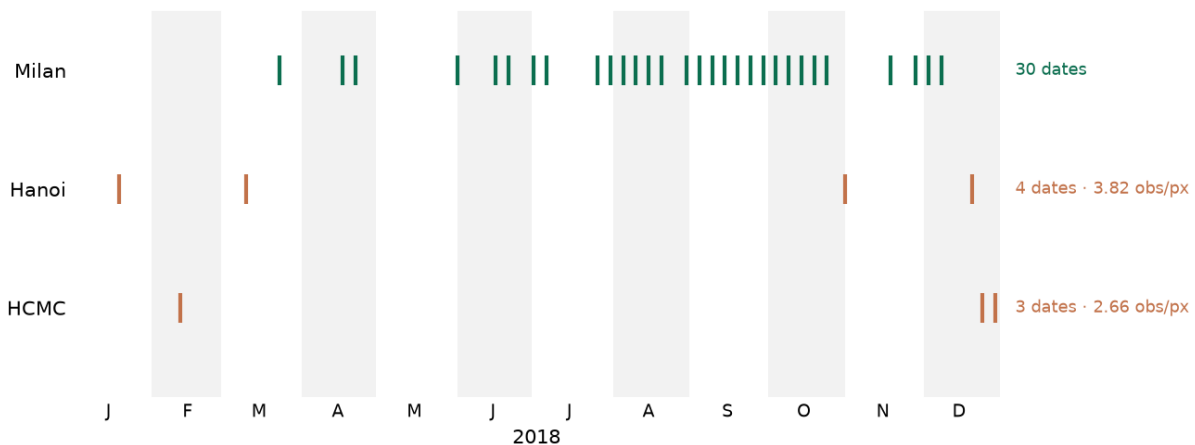


Figure 1: Usable Sentinel-2 acquisitions per city after cloud screening, 2018. Each tick is one retained date. Milan's 30 dates carry an average of 30 observations per pixel, Hanoi's 4 carry 3.82 and Ho Chi Minh City's 3 carry 2.66. Ho Chi Minh City's two December dates are five days apart and are one effective look rather than two.

Three composite methods are defined over whatever dates survive. The median composite reduces the stack per band to its median, giving 10 bands. The stack composite takes four dates chosen one per season and concatenates them, giving 40 bands. The percentile composite reduces the stack to five percentiles per band, p10, p25, p50, p75 and p90, giving 50 bands.

Only the first is computable in Vietnam. A percentile is only meaningful once enough observations sit on either side of it, and the guard used here scales with the number of percentiles requested: five percentiles require at least 17 dates. Hanoi's 4 and Ho Chi Minh City's 3 fall far short.

The shortfall is not a consequence of the screening thresholds, and this was tested rather than assumed.

The scene search evaluates a ceiling case in which every threshold is disabled and every candidate acquisition is accepted regardless of cloud, validity or coverage.

Even then Ho Chi Minh City reaches 16 dates against the 17 required. It fails by one. Hanoi reaches 10 and fails by seven. Neither city can compute a percentile composite from imagery that does not exist, and Ho Chi Minh City is the sharper case precisely because it comes so close: one further acquisition anywhere in 2018 would have carried it over the threshold, and no adjustment to the screening can conjure one, because at the ceiling nothing is being screened out at all. The 16 dates are the whole of what the 2018 archive holds over that area of interest.

The narrowness of that margin does not weaken the conclusion, and it is worth saying why. The ceiling set is not a usable alternative that was passed over: it consists of every acquisition including those rejected for cloud, which is how a count of 16 becomes a screened count of 3. A percentile composite computed from it would be a percentile of largely cloudy observations.

The stack composite fails for the same reason at a smaller count, since it requires four dates spread one per season and Ho Chi Minh City's three fall in two months.

The median composite has no minimum date count, so it was the only method that could be computed in Vietnam at all. This is an archive limitation and not a modelling preference, and it should not be read as a finding that median is the better choice in Vietnam: Section 4 shows it is the weakest of the three in Milan, where all three could be computed.

Two further properties of these composites are recorded here because later sections rest on them.

Milan's composite is near-annual rather than annual. January, February and May are absent from all 30 dates, so the composite leans towards the growing season. Describing it as annual would overstate its coverage.

Ho Chi Minh City's composite is thinner than a count of three suggests. Its three dates are 12 February, 24 December and 29 December. The two December acquisitions are five days apart and sample the same scene state, so the composite is effectively one February look plus one late-December look, spanning two months of twelve. Hanoi's four dates spread across four months. The consequence is that the Milan median draws on about 30 looks per pixel while the Vietnamese medians draw on between two and four, and Section 8 records what that confounding forbids.

2.4 Photo-interpreted plots

The validation reference is a set of 450 photo-interpreted plots per city, 1 350 in total, produced by Keerthana Kirubakaran in Google Earth Pro on 2018 imagery and labelled with EarthLabel. It is independent of every product above: no model saw it at any stage, and it is used for validation alone.

A plot is a 10 m unit area, which is exactly one Sentinel-2 pixel and one pixel of every map under test. Each plot is subdivided into a three-by-three grid of nine sub-cells, and each sub-cell is interpreted individually and assigned a land-cover code. The plot's IMD is then the percentage of its sub-cells that are impervious. The nine sub-cells are never observations in their own right: they are collapsed to one value per plot before anything is scored, so every metric in Section 6 runs on 450 observations per city. One code, `Rock/material`, is ambiguous between sealed and unsealed and is dropped wherever it occurs, with the denominator renormalised so that a plot with one dropped sub-cell scores out of eight rather than being discarded. It occurs once, in Milan.

Which codes count as impervious is a definitional choice rather than an observation, so it is not fixed. Three rules are carried through the whole of Section 6 and never averaged or collapsed:

- **strict**, the primary rule, counts parking, pavement, road, roof, railroad, airport and pool as impervious;

- **B** adds permeable pavement, which matches the sealed-surface definition CLMS works to;
- **C** adds unpaved dirt road and heavily compacted bare soil, on the ground that compacted surfaces behave hydrologically much as sealed ones do.

Paved roads and railroads are impervious under all three rules, including the primary one. Only dirt roads move between rules, and only under C. This is worth stating precisely, because it is what makes the GHS-BUILT-S comparison in Section 6 a fair one: the reference counts paved roads as impervious under every rule while the target excludes them by design, and no choice of rule removes that mismatch.

The two additional codes are rare, which bounds how much the rule choice can matter. In Milan permeable pavement occurs in 23 sub-cells across 6 plots and dirt road in 112 sub-cells across 18 plots. In Hanoi and Ho Chi Minh City permeable pavement does not occur at all, so rule B is identical to strict there by construction rather than by coincidence; dirt road occurs in 63 sub-cells across 13 plots in Hanoi and 34 across 10 in Ho Chi Minh City. Section 6.3 reports the sensitivity of the results to the choice.

3 Methods

3.1 Sampling, spatial blocking and buffering

The Milan point set is Žgela’s and is used unchanged. His design is summarised here rather than re-derived; the provenance and the rationale are his [1, pp. 2, 4, Figs. 1–2].

The CLMS reference map was classified into seven IMD groups (0 %, 1 to 20, 21 to 40, 41 to 60, 61 to 80, 81 to 99 and 100 %) and 500 points were drawn at random within each group, giving 3 500 points in total. Stratifying this way rather than sampling uniformly is what puts enough points in the sparse middle of the range, which an unstratified draw over a city would leave nearly empty.

Train and test are then separated spatially rather than at random. The area is divided into 1 km by 1 km blocks and each block is assigned whole to either training or testing, so no test point shares a block with a training point. A 250 m buffer is applied on top, removing test points that fall within 250 m of any training point. What remains is 2 449 training points and 1 014 test points, approximately a 70/30 split. Random splitting would place training and test points metres apart in the same neighbourhood, where they share land cover and often the same rooftops, and would return an accuracy figure that measures interpolation between neighbouring pixels rather than prediction.

The design’s own spatial randomness was checked by Žgela with an average nearest neighbour index [7], which runs from 1.02 for the 0 % class down to 0.75 for the 100 % class. The lower value records that fully impervious pixels cluster in the city centre, which he acknowledges as a limitation of the sampling rather than correcting for it. It is repeated here on the same terms.

Two consequences for this report follow. The first is that the 1 014-point holdout is the same holdout throughout Section 4, and the count is not a coincidence of four separate splits agreeing: notebook 01b imports the split from the embeddings run rather than recomputing it, so the four Milan predictor sets are scored on exactly the same held-out points. That is what makes their comparison a comparison of predictors. The second is that the 1 014 figure matches the one Žgela reports, which is one of the checks that the shared baseline was correctly reconstructed. Figure 2 shows the resulting split.

Vietnam is sampled and split on the same design, against GHS-BUILT-S in place of CLMS, giving 895 test points in Hanoi and 887 in Ho Chi Minh City.

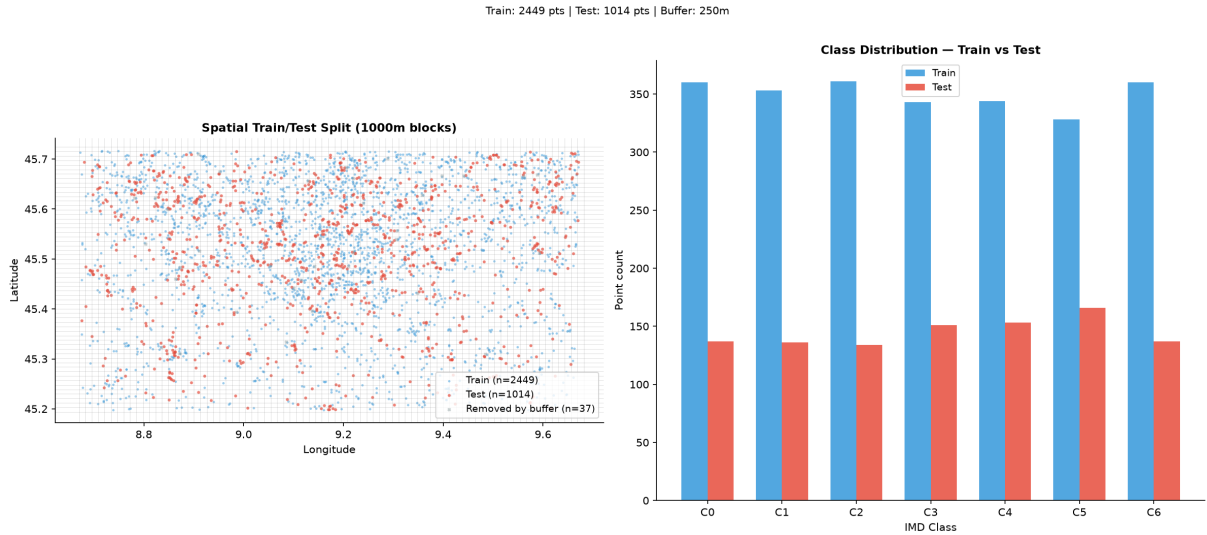


Figure 2: Training points, test points and buffer-removed points over the 1 km block grid, Milan, with the class counts alongside. Blocks are assigned whole to training or testing and test points within 250 m of a training point are removed.

3.2 Estimators, tuning and spatial cross-validation

Three estimators were tuned on the Milan data under a randomised hyperparameter search with spatially aware five-fold cross-validation, at three block sizes: 500 m, 1 km and 2 km. Two are carried forward and reported here, a random forest [8] and a support vector regressor [9]. Tuning is performed with scikit-learn [10]. Only estimators whose trained form can be applied to a raster through the Google Earth Engine Python API [11] were considered for the final maps, since every map in this report is produced by predicting over the full scene in Earth Engine rather than by scoring a table of points.

The scope of each estimator is fixed once, here, and holds for the rest of the report. SVR is evaluated in Milan only. Every model outside Section 4 is a random forest. Two reasons make that necessary rather than convenient. The transfer experiment of Section 5 requires a random forest by construction, so the Milan model carried to Hanoi and Ho Chi Minh City must be one. And the comparison the report is built on, across predictor sets, across scenarios and across cities, is only a comparison of those things if the estimator is held fixed, since a difference between a random forest map and an SVR map cannot be attributed to the predictor set. The independent validation of Section 6 likewise scores the random forest raster from every run for the same reason.

Cross-validation is spatial throughout, using the same block structure as the train/test split, so a fold boundary is a spatial boundary. The point of doing so is to prevent the tuning score from being inflated by spatial autocorrelation, and whether it worked is measurable: Figure 3 compares cross-validation RMSE under spatial folds against random folds, per estimator and per block size.

The differences are negligible. The largest absolute inflation across the reported estimators and all three block sizes is 0.3 %, and most cells are smaller still. Spatial and random cross-validation agree to within about one per cent, which says that the buffering and blocking of Section 3.1 have already removed what leakage there was to remove, and that the tuning scores are not inflated by proximity between folds. It does not say that spatial blocking was unnecessary: the reason the two agree is that the split was built to make them agree.

Cross-validation scores are used for tuning and are never quoted as holdout performance. They score scikit-learn models on training-set folds, whereas every accuracy figure in Sections 4 to 6 is scored on data held out from training. Section 4.2 takes up the one place where the two disagree in direction.

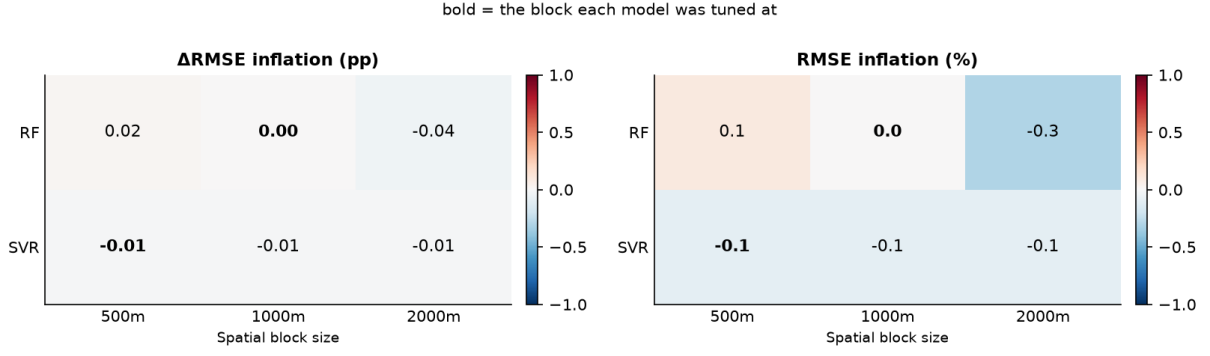


Figure 3: Cross-validation RMSE under spatial blocking against random folds, Milan training set, by estimator and block size. Bold marks the block size each estimator was tuned at. Values are the difference in RMSE, in percentage points on the left and as a percentage of the random-fold score on the right.

3.3 Accuracy metrics

Four metrics are reported throughout, and a fifth is derived from RMSE in the independent validation. For n observations with reference values y_i and predicted values $\hat{y}_i$, and with $\bar{y}$ the mean of the reference values:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

the root mean squared error, in IMD percentage points;

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

the mean absolute error, in the same units;

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (3)$$

the proportion of reference variance the map accounts for;

$$\text{Bias} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i) \quad (4)$$

the mean signed error, in the same units.

RMSE and MAE differ in how they weight the error distribution. MAE averages the absolute errors, so every plot contributes in proportion to how wrong it is. RMSE squares first, so it is driven by the largest errors and a map with a few gross failures scores worse on RMSE than on MAE. The two can therefore disagree about which of two maps is better, and Section 6.1 contains a case where they do. Both are reported rather than one, and neither is treated as the summary of the other.

Bias is observed minus predicted, reference minus map. The sign convention is stated here because it is the one thing in this section that a reader can invert without noticing:

- a **positive** bias means the reference is higher than the map, so the map **under-predicts**;

- a **negative** bias means the map reads higher than the reference, so the map **over-predicts**.

The convention is identical in every notebook that produces a number in this report, and nothing was harmonised after the fact: the definitions already agreed. It matters because the two validations produce biases of opposite sign and both are correct. Section 5 reports negative biases in Vietnam, where the models read higher than GHS-BUILT-S. Section 6 reports positive biases for GHS-BUILT-S itself, where the product reads lower than the interpreted reference. Those are different comparisons, not a contradiction, and the signs must not be read across them.

The fifth metric is reported alongside RMSE in the independent validation. The photo-interpreted reference is itself measured with error: a plot's IMD is the proportion of nine interpreted sub-cells that are impervious, so it is a proportion estimated from nine draws and carries a sampling error of its own. Writing n_i for the number of sub-cells retained at plot i , normally nine but eight where an ambiguous cell was dropped and the denominator renormalised, and taking the predicted value $\hat{y}_i$ as a proportion for the plug-in estimate of the plot's true impervious fraction, the binomial variance of the reference estimate is $\hat{y}_i(1 - \hat{y}_i)/n_i$. Averaging it across plots and removing it from the mean squared error gives:

$$\text{RMSE}_{\text{corr}} = \sqrt{\max\left(0, \text{RMSE}^2 - \frac{1}{n} \sum_{i=1}^n \frac{\hat{y}_i(1 - \hat{y}_i)}{n_i}\right)} \quad (5)$$

the noise-corrected RMSE, in the same units: an estimate of map error with the reference's own measurement error removed.

The correction is floored at zero because it is a difference of two variances and can go negative for a map that is closer to the reference than the reference's own noise.

$\text{RMSE}_{\text{corr}}$ is an **upper bound** on map error rather than a point estimate, and is used only as one. The independence assumption behind the binomial correction does not hold: nine sub-cells within a single 10 m pixel share land cover and are spatially correlated, so the binomial calculation understates the true reference noise and the correction consequently removes less error than it should. The true map error is therefore at or below $\text{RMSE}_{\text{corr}}$, and the correct use of the figure is to say that a map's error is at most some value, never that it equals it.

3.4 The two-validation design

Every map in this report is scored twice, against two different references, and the distinction between them is the methodological hinge of the whole study.

Same-source validation scores each map against the product it was trained on: CLMS in Milan, GHS-BUILT-S in Hanoi and Ho Chi Minh City. It runs on the spatial holdout described in Section 3.1, 1 014 points in Milan, 895 in Hanoi, 887 in Ho Chi Minh City, which is held out from training but drawn from the same product as the training labels. What it measures is agreement with the training target. It is reported in Sections 4 and 5.

Independent validation scores every map against the photo-interpreted plots of Section 2.4, 450 per city, which no model saw at any stage and which were produced from imagery rather than from either target product. What it measures is accuracy against an independent reference. It is reported in Section 6.

Both are needed, and neither substitutes for the other. Same-source validation is available for every candidate model, uses the full spatial holdout, and is the natural instrument for choosing among models. What it cannot do is distinguish a good map from one that merely resembles its labels. A model scored against the product it was fitted to is rewarded for reproducing that product, including wherever the product is wrong, and a map that reproduces a systematic error in its target scores well for doing so.

That limitation is not a hypothetical here: Section 6 finds that both training products, scored against photo-interpretation on the same terms as the models, do no better than the models fitted to them.

Independent validation answers the question same-source validation cannot, at the cost of a much smaller sample, a stratified rather than area-weighted one, and a reference carrying interpretation error of its own. Section 8 records those costs.

The two are never merged and their numbers are never compared directly. They are scored against different references, on different samples, at different sizes, and a difference between a same-source number and an independent one would measure the change of reference rather than anything about a map. Every table in this report belongs to one validation or the other and says which. The one place the two are set side by side is Section 7.1, which is about the relationship between them and treats the change from one to the other as its subject rather than reading across it.

4 Milan results — same-source validation

All three Sentinel-2 composites outperform the AlphaEarth embeddings against CLMS. The percentile composite reaches an RMSE of 9.462 against the embeddings' 14.123, a reduction of 33 %.

Three estimators were tuned under the same randomised search and spatial block cross-validation; the two carried forward, random forest and SVR, are the ones reported here. SVR appears in this section only. Every map outside Section 4 is a random forest, because the transfer design in Section 5 requires one and the independent validation holds the estimator fixed across runs so that the comparison isolates the predictor set rather than confounding predictor with estimator.

The numbers in this section come from a single validation: each Milan map scored against CLMS, the product it was trained on, over the 1 014-point spatially buffered holdout. They measure agreement with the training target and not accuracy. Results against the photo-interpreted plots are an independent validation and are reported in Section 6; the two are never compared directly. In particular, the paired per-plot testing of these same four predictor sets is independent validation work and belongs to Section 6.2.

4.1 Four predictor sets ranked

[Table 1](#) gives the holdout metrics for the four Milan predictor sets under the GEE random forest.

Table 1: Milan holdout metrics against CLMS, GEE random forest, $n = 1\,014$. Bias is observed minus predicted, so a positive value means the map under-predicts.

Predictor set	RMSE	MAE	R^2	Bias
S2 percentile (50 bands)	9.462	6.389	0.927	-0.104
S2 stack (40 bands)	10.864	7.641	0.904	-0.169
S2 median (10 bands)	11.285	7.820	0.896	0.046
AlphaEarth embeddings (64 bands)	14.123	10.675	0.837	0.624

The ordering is the same on RMSE, on MAE and on R^2 , so it is not an artefact of a single metric. [Figure 5](#) shows the three metrics side by side, which is the form in which that claim can be checked rather than taken on trust: the four bars fall in the same order in all three panels. The gap between the best composite and the embeddings, 4.661 RMSE, is larger than the gap between the best and worst composite, 1.823. All four maps are close to unbiased on this validation, between -0.169 and 0.624, which is expected of a model fitted to the product it is then scored against.

The equivalent SVR rows follow the same ordering: percentile 9.051, stack 10.219, median 10.939, embeddings 14.756. The advantage of the composites over the embeddings therefore does not depend on the estimator.

Figure 4 shows observed against predicted values on the holdout for the percentile composite, and Fig. 6 breaks the same run down by IMD class.

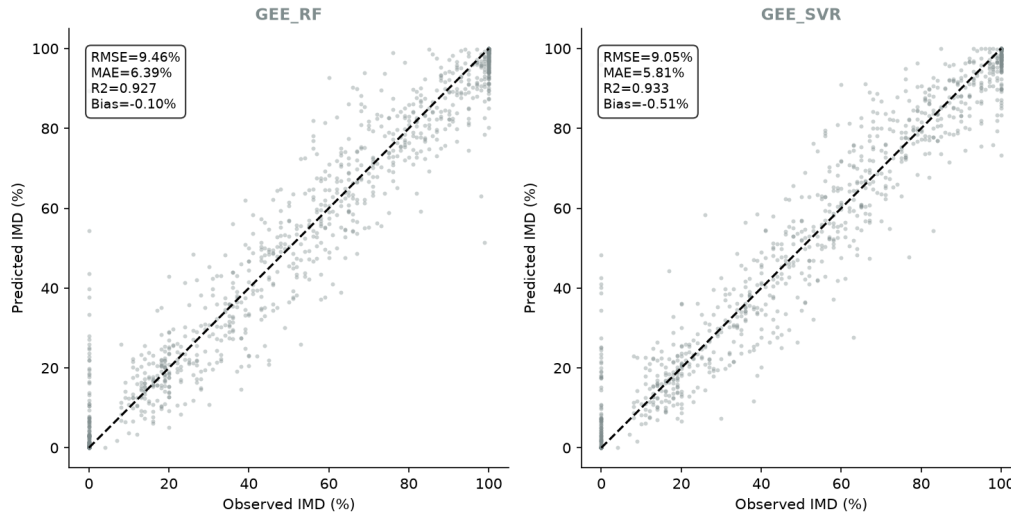


Figure 4: Observed CLMS against predicted IMD on the 1 014-point holdout, S2 percentile composite, random forest and SVR panels.

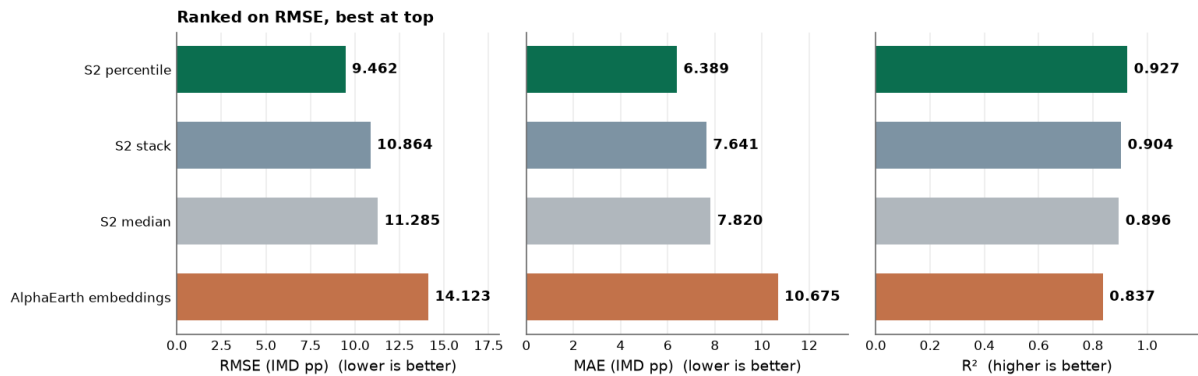


Figure 5: The four Milan predictor sets on the 1 014-point spatial holdout, scored against CLMS under the GEE random forest, ranked on RMSE with the best at the top. The same four values appear in Table 1. Bars start at zero on all three panels, so bar length is proportional to the metric and the spread is not exaggerated by a truncated axis. Section 7.1 returns to that spread, which compresses sharply under the independent validation.

Figure 7 maps the percentile prediction against CLMS across the whole scene. The two upper panels agree on the structure of the conurbation: the dense core, the satellite towns to the north, and the largely agricultural south. The difference panel is where the disagreement is legible. It is close to zero over most of the rural area and over the interior of the built-up core, and departs from CLMS in two places. The first is a fine network of positive lines threading the countryside, following the road and canal network: the model reads these narrow sealed features as more impervious than CLMS does. The second is a mild negative cast over the densest part of the city centre, where the model sits below CLMS. The predicted panel also carries small unmapped gaps along the watercourses that CLMS fills. The map is qualitative support for Table 1 and not a measurement: the whole-scene comparison is not the 1 014-point holdout the metrics are scored on, and no number is quoted from it.

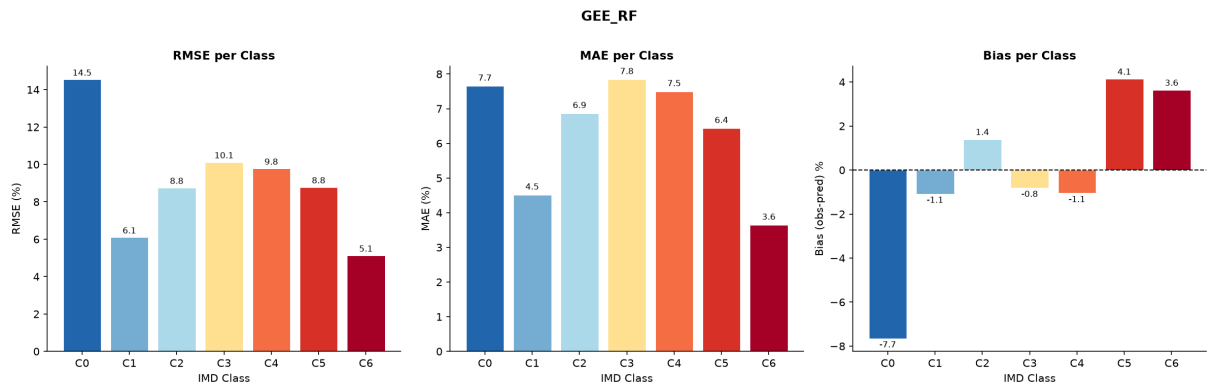


Figure 6: RMSE, MAE and bias per IMD class, S2 percentile composite, GEE random forest, scored against CLMS.

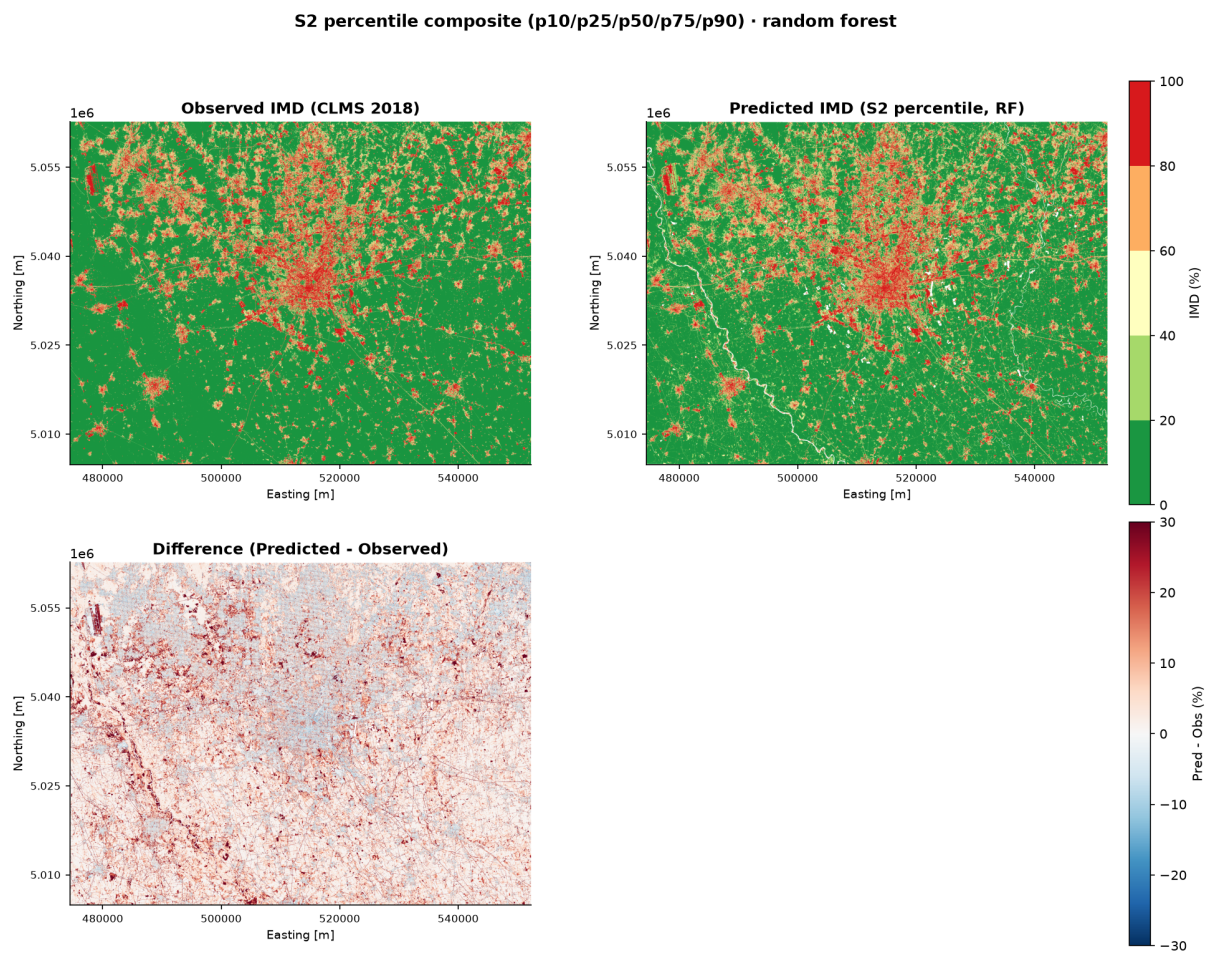


Figure 7: Milan IMD, observed CLMS against the predicted map and their difference, S2 percentile composite, random forest. Positive difference means the model reads more impervious than CLMS.

4.2 The CV-versus-holdout reversal

SVR wins cross-validation and loses the holdout. This reproduces Žgela’s result [1, p. 5] rather than adding to it: he reports SVR best on cross-validation with RF close behind, then RF winning every holdout metric, and selects RF for that reason.

The same reversal appears here. On the embeddings run SVR is tuned to a cross-validation RMSE of 11.711 at its selected 500 m block, against RF’s 13.162 at 1 000 m, so SVR leads by 1.451 on cross-validation. It then loses the holdout on all three of RMSE, MAE and R^2 : 14.756 against 14.123 on RMSE, 11.400 against 10.675 on MAE, and 0.822 against 0.837 on R^2 . The cross-validation values are scores on training-set folds and are not comparable with the holdout figures in Table 1; only the direction of each comparison is being read across them.

The reversal carries a confound that should be stated. The holdout rows are scored on rasters built in Earth Engine by `smileRandomForest` and `libsvm`, while the cross-validation numbers come from the tuned scikit-learn models, and the parameter translation between the two is lossy in both directions. The reversal may therefore reflect overfitting to the cross-validation folds, or it may reflect how faithfully each Earth Engine estimator reproduces its scikit-learn counterpart. The two cannot be separated from these runs. Consistently with this, the embeddings run records SVR as its cross-validation winner in metadata, while RF is the model carried downstream.

RF is carried forward because the transfer experiment in Section 5 requires a random forest and the comparison across cities holds the estimator fixed (Section 3.2). It is not carried forward because SVR could not be rastered: SVR is supported in Earth Engine through `ee.Classifier.libsvm` [12] with `svmType='EPSILON_SVR'`, and the SVR raster for Milan was produced and retained.

4.3 Where the signal is

The percentile composite’s advantage comes from the low percentiles of the red band and the high percentiles of the near infrared, a seasonal signal that a single median cannot express.

Table 2 gives the five most important bands of the percentile run by permutation importance on the holdout, with mean decrease in impurity alongside, and Fig. 8 gives the ranking across all fifty bands.

Table 2: Top five bands by permutation importance, S2 percentile composite, random forest, Milan holdout.

Rank	Band	Permutation importance	Impurity importance
1	B4_p25	0.1812	0.3518
2	B4_p10	0.0556	0.1163
3	B8_p90	0.0401	0.0664
4	B4_p50	0.0225	0.0751
5	B8_p75	0.0211	0.0446

B4_p25 alone carries more than three times the permutation importance of the next band. Four of the top five are low percentiles of red or high percentiles of near infrared, which are precisely the quantiles a median composite discards. That is a mechanistic explanation of the ranking in Section 4.1: the percentile composite does not simply carry more bands than the median, it carries the particular bands the median throws away.

The comparison with the embeddings run is instructive. There the two most important predictors are bands B16 and B08 of the 64-dimensional embedding [1, p. 6, Fig. 7], which have no physical interpretation. The Sentinel-2 composites give an interpretable answer to why they work; the embeddings do not.

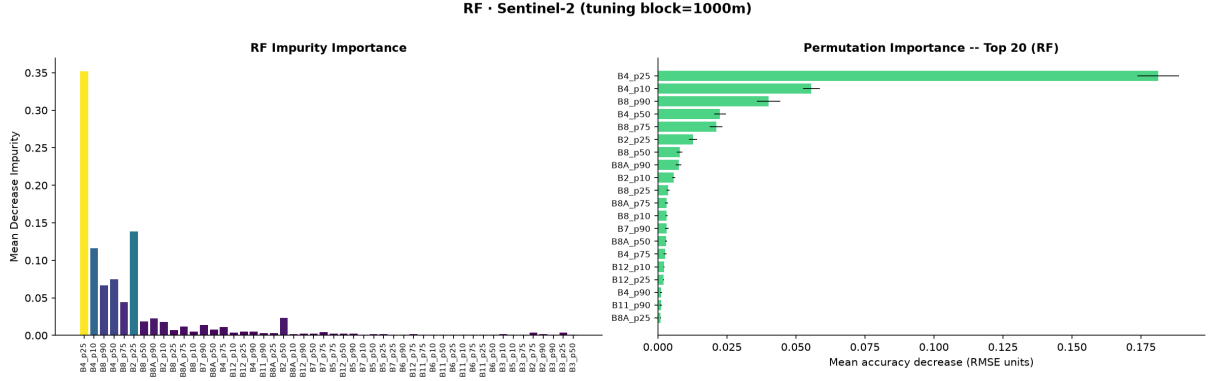


Figure 8: Impurity and permutation importance per band, random forest, S2 percentile composite.

5 Vietnam results — same-source validation

Zero-shot transfer of a Milan-trained model to Hanoi or to Ho Chi Minh City fails. Local retraining is both necessary and sufficient to recover usable accuracy.

Every model in this section is a random forest. SVR is not carried outside Milan: the transfer design requires a random forest, and holding the estimator fixed is what makes the comparison across cities and predictor sets a comparison of predictors rather than of estimators.

As in Section 4, these are same-source numbers. Each Vietnamese map is scored against GHS-BUILT-S, the product the local models were trained on, over that city’s spatial holdout. They measure agreement with the training target. Section 6 scores the same maps against photo-interpretation, and the two sets of numbers are not compared.

5.1 Zero-shot versus local retrain

Table 3 gives both scenarios for both predictor sets in both cities.

Table 3: Vietnam holdout metrics against GHS-BUILT-S, random forest. Bias is observed minus predicted, so a negative value means the map over-predicts.

City	Predictor set	Scenario	RMSE	MAE	R^2	Bias	n
Hanoi	AlphaEarth embeddings	zero-shot	35.960	30.140	−0.070	−24.930	895
Hanoi	AlphaEarth embeddings	local retrain	23.860	17.720	0.529	−0.690	895
Hanoi	S2 median	zero-shot	37.150	28.930	−0.142	−23.790	895
Hanoi	S2 median	local retrain	24.050	18.490	0.522	−2.080	895
HCMC	AlphaEarth embeddings	zero-shot	40.650	34.170	−0.279	−29.760	887
HCMC	AlphaEarth embeddings	local retrain	21.310	16.580	0.649	0.400	887
HCMC	S2 median	zero-shot	35.190	26.980	0.042	−23.180	887
HCMC	S2 median	local retrain	22.880	17.930	0.595	−1.900	887

The zero-shot R^2 is at or below zero in three of the four city and predictor combinations: −0.070 and −0.142 in Hanoi, −0.279 for the embeddings in HCMC. A model with R^2 at or below zero is no more informative than predicting the mean of the reference everywhere. The fourth combination, the S2 median in HCMC at 0.042, is not a counter-example so much as a weaker instance of the same failure.

Local retraining recovers R^2 to between 0.522 and 0.649 in every case, and cuts RMSE by roughly a third to a half: from 35.960 to 23.860 and 37.150 to 24.050 in Hanoi, and from 40.650 to 21.310 and 35.190 to 22.880 in HCMC. The improvement holds for both predictor sets and in both cities, which is

what makes the conclusion a statement about transfer rather than about a particular feature space.

The bias column identifies the failure mode. Every zero-shot bias is large and negative, between -23.180 and -29.760 , so the transferred models read systematically more impervious than GHSL across the whole test set. They are not scattered around the reference; they sit above it. The local retrains bring bias to between -2.080 and 0.400 , near zero on all four. Section 7.2 takes up what this level mismatch is and why it arises.

Both predictor sets fail in the same way and to a similar degree, so the failure is not a property of the AlphaEarth embeddings or of the Sentinel-2 composites specifically. One caveat belongs with that statement and is developed in Section 8: the two zero-shot experiments are not equally handicapped, because the embeddings compare annual coverage against annual coverage while the S2 median compares Milan's 30-date composite against 3 to 4 Vietnamese dates. Composite depth is confounded with predictor type here, so the near-equality of the two failures should not be read as a controlled comparison between them.

Figures 9 and 10 show the two predictor sets separately, on identical axes. They are the same experiment run on different features, so they are read as a pair: the shape of the result is the same in both, which is the point of Section 5.1.

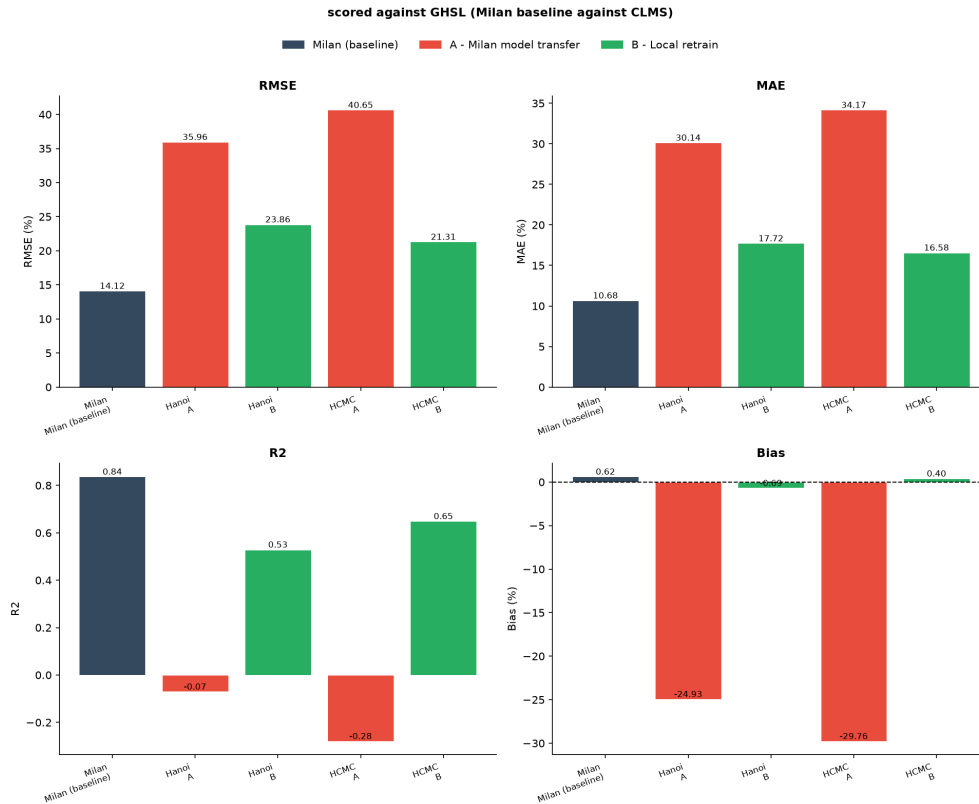


Figure 9: AlphaEarth embeddings. Milan baseline, zero-shot transfer and local retrain across all four metrics, Hanoi and HCMC. Every bar is scored against the training target, GHS-BUILT-S in Hanoi and HCMC and CLMS for the Milan baseline, and measures agreement with that target rather than accuracy. In particular a bias bar near zero indicates agreement with GHSL, not a correct map.

Figure 11 shows the same three maps as rasters, where the level shift the metrics report is visible directly.

5.2 Per-class behaviour

Breaking the same holdout down by IMD class shows that local retraining is not uniformly better, and locates both where it helps and where it costs.

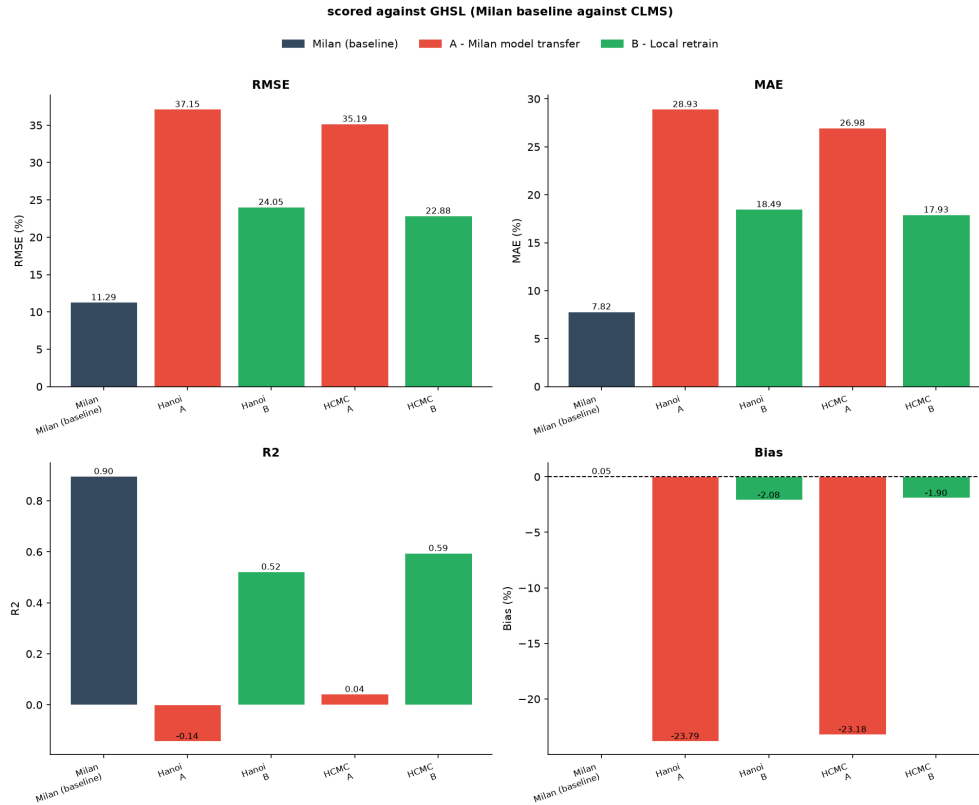


Figure 10: S2 median composite. The same four metrics and the same two scenarios as Fig. 9, for the Sentinel-2 median predictor set rather than the embeddings, and scored against the same targets: GHS-BUILT-S in Hanoi and HCMC, CLMS for the Milan baseline. Again agreement with the target, not accuracy.

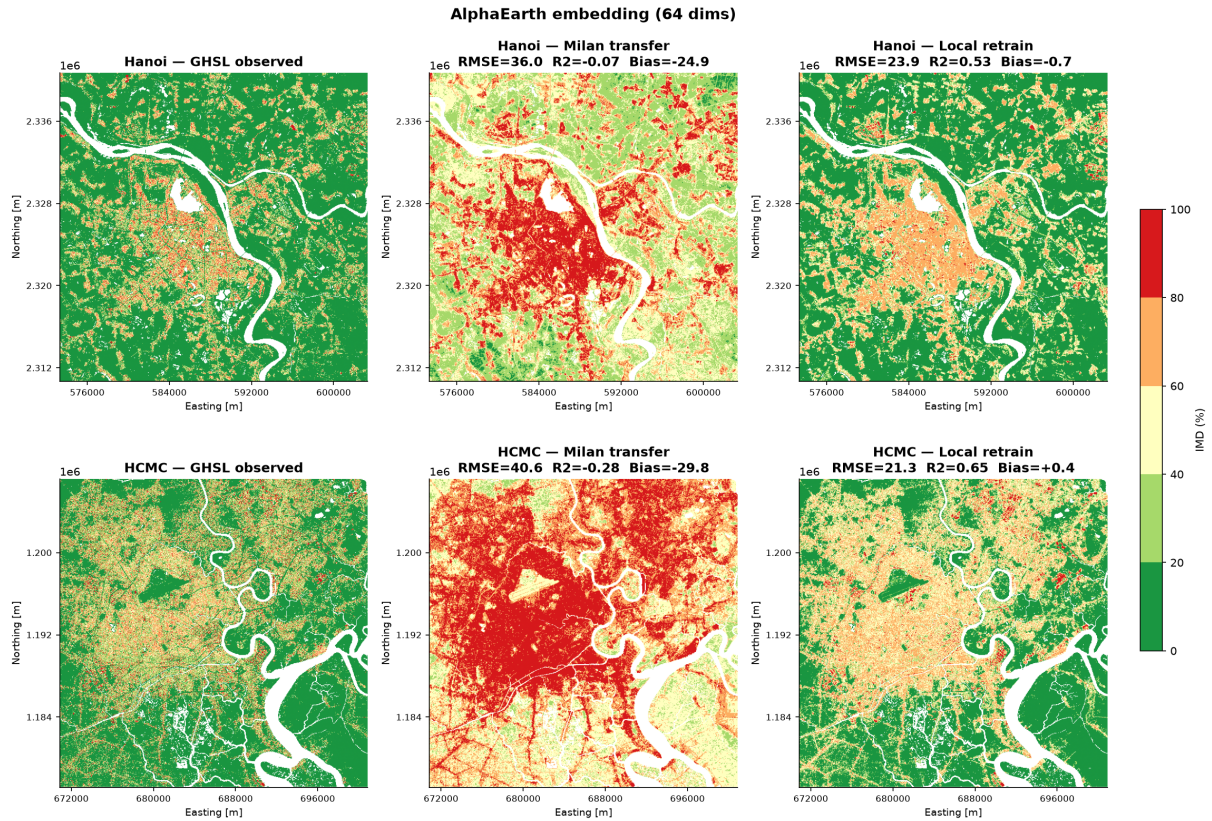


Figure 11: GHS-BUILT-S, the Milan zero-shot transfer and the local retrain as rasters, Hanoi and HCMC, AlphaEarth embeddings, scored against GHS-BUILT-S.

Retraining helps sharply in the low and middle classes. In Hanoi the embeddings model’s MAE in class C0 falls from 40.75 to 6.36, and in C1 from 50.08 to 20.12; in HCMC C0 falls from 49.05 to 5.15. These are the classes where the transferred model’s high level does most damage, since a map that cannot predict low values is worst where the reference is lowest.

Above 80 % imperviousness the ordering reverses. In Hanoi the embeddings model’s C6 MAE rises from 17.25 under zero-shot to 30.95 after retraining, and in HCMC from 12.93 to 25.20. The S2 median behaves the same way: Hanoi C6 rises from 8.07 to 26.16 and HCMC C6 from 5.91 to 22.58. This is a real limitation of local retraining and not noise. Žgela reports the same pattern [1, p. 9], with improvements mainly in classes C0 to C3 and the retrained model challenged at the top of the range.

The mechanism is visible in the same rows. The transferred models over-predict everywhere, which is ruinous in the low classes and incidentally close to right in the highest one, where the reference is near 100 % and a map biased upward has little room to err. Retraining removes the upward bias globally, which fixes the low classes and removes the accident that was flattering the top class.

Per-class R^2 is available but is not quoted here. Restricting to a single IMD class removes most of the variance that R^2 is normalised by, so every defined per-class value is negative even where the same model reaches a global R^2 of 0.522 to 0.649. That is arithmetic rather than model failure. Classes C0 and C6 are single-valued strata, so their variance is exactly zero and their R^2 is undefined. Per-class RMSE, MAE and bias are the appropriate within-class measures, and R^2 is reserved for the global comparison in Section 5.1.

Figure 12 shows the pattern across all seven classes.

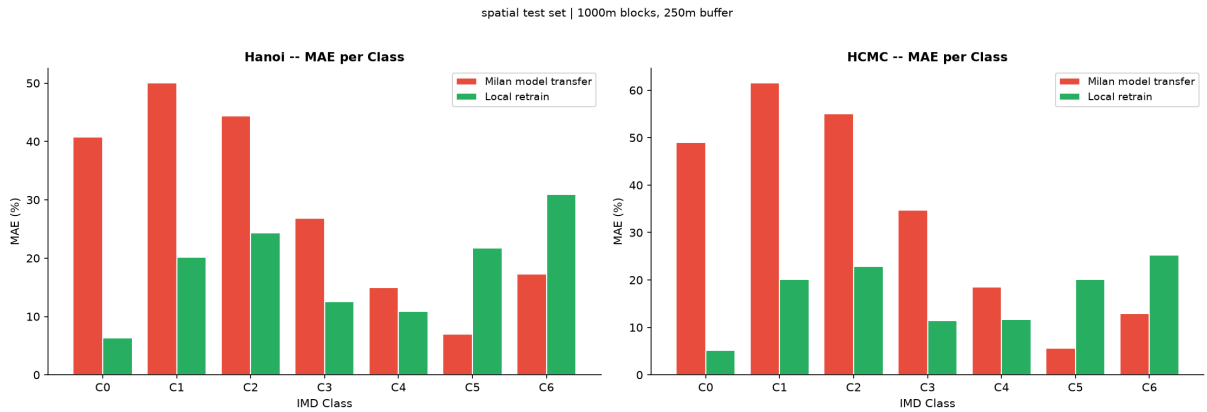


Figure 12: MAE per IMD class, zero-shot transfer against local retrain, Hanoi and HCMC, AlphaEarth embeddings, scored against GHS-BUILT-S. The S2 median figures quoted in this subsection are read from that run’s per-class table rather than from a second copy of this figure.

6 Independent validation

Scored against photo-interpretation, the Milan spread that Section 4 reports largely disappears, and the two training products score no better than the models fitted to them.

This section is a second, independent validation and not a continuation of the first. Every map in all three cities, models and training products alike, is scored on 450 photo-interpreted plots per city that no model saw at any stage. The reference is the same for every map in a city, which is what allows CLMS and GHS-BUILT-S to appear here on the same terms as the models: in this validation they are maps under test, not targets. As stated in Section 3.4, the two validations are never merged, and no number in this section is compared with a number from Section 4 or Section 5.

6.1 All fifteen maps

Table 4 gives every map under the primary strict rule.

Table 4: All fifteen maps against 450 photo-interpreted plots per city, strict rule. One plot is a 10 m unit area equal to one pixel, and plot IMD is the percentage of its nine interpreted sub-cells that are impervious. Bias is observed minus predicted, so a positive value means the map under-predicts the interpreted reference. $\text{RMSE}_{\text{corr}}$ is the binomial reference-noise correction and is an upper bound on map error rather than a point estimate.

City	Map	Role	RMSE	$\text{RMSE}_{\text{corr}}$	MAE	R^2	Bias
Milan	S2 stack	model	24.644	21.559	16.551	0.652	-3.344
Milan	S2 percentile	model	24.809	21.874	16.486	0.647	-3.342
Milan	emb_RF	model	25.625	22.420	18.299	0.624	-1.622
Milan	S2 median	model	25.984	22.917	18.363	0.613	-4.176
Milan	CLMS	reference	26.253	24.411	14.606	0.605	2.263
Hanoi	emb_localrf	model	26.502	23.266	19.588	0.649	6.592
Hanoi	S2_median_localrf	model	27.282	23.885	21.320	0.628	8.181
Hanoi	S2_median_zeroshot	model	33.791	31.195	24.469	0.429	-15.876
Hanoi	emb_zeroshot	model	34.352	31.061	29.537	0.410	-15.716
Hanoi	GHSL	reference	40.345	38.833	27.052	0.187	19.680
HCMC	S2_median_zeroshot	model	25.950	23.420	16.779	0.665	-6.827
HCMC	emb_localrf	model	27.242	24.011	20.294	0.630	11.088
HCMC	S2_median_localrf	model	29.615	26.484	23.030	0.563	12.233
HCMC	GHSL	reference	37.355	35.597	25.730	0.305	19.800
HCMC	emb_zeroshot	model	38.494	35.775	32.404	0.262	-19.641

The Milan result is the compression. The four predictor sets span 24.644 to 25.984 RMSE, a range of 1.340. Their ordering is preserved from the same-source validation only in part: S2 stack now leads on RMSE at 24.644 with S2 percentile 0.165 behind it at 24.809, and the AlphaEarth embeddings at 25.625 are no longer the worst map, having been passed by the S2 median at 25.984. Section 7.1 takes up what this compression means for model selection.

In Vietnam the spread is wider and its structure is different, because the maps being compared differ in kind rather than only in predictor set. In both cities the two local retrains sit well ahead of both zero-shot maps, at 26.502 and 27.282 in Hanoi against 33.791 and 34.352, which reproduces the Section 5 conclusion against a reference the models never saw. HCMC is the exception that Section 7.2 accounts for: the S2 median zero-shot map records the city’s best RMSE at 25.950, ahead of both local retrains. That is a consequence of where the city’s reference level happens to fall relative to the transferred map’s level and not evidence that transfer succeeded there. The embeddings zero-shot map in the same city is the worst of all fifteen at 38.494.

Figure 13 is the summary view, and carries the bootstrap confidence intervals that the table omits. The Milan intervals overlap one another substantially, and the Milan models’ intervals overlap CLMS’s.

The diamonds in that figure carry a result of their own. Both training products score no better than the models fitted to them. CLMS reaches an RMSE of 26.253 in Milan, worse than all four of the Milan models it trained, the closest of which is the S2 median at 25.984. GHS-BUILT-S reaches 40.345 in Hanoi and 37.355 in HCMC, worse than every local retrain in either city and, in Hanoi, worse than every map of any kind. Its bias is +19.680 in Hanoi and +19.800 in HCMC, so it under-marks the interpreted reference by close to 20 percentage points in both cities. That is the road exclusion of Section 2.1 appearing as a number.

One apparent counter-example needs settling, because CLMS holds the best MAE of any Milan map at 14.606 alongside the worst RMSE at 26.253. Table 5 gives the per-plot error distribution behind that split.

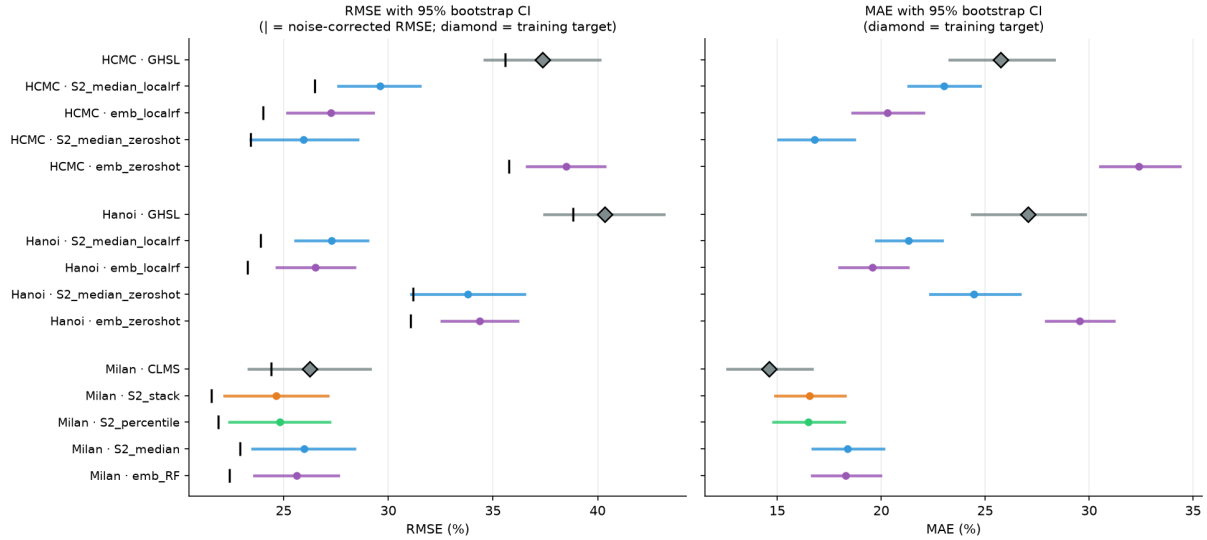


Figure 13: RMSE and MAE with 95 % percentile bootstrap [13] confidence intervals for all fifteen maps, three cities, strict rule. Intervals are over 10 000 resamples of the 450 plots, the plot being the independent unit. Diamonds mark the training products, scored here as maps rather than as targets; the tick on each RMSE bar is the noise-corrected RMSE.

CLMS is right far more often than any model, landing within 5 percentage points of the interpretation on 53.3 % of plots against 34.2 % for the closest model, and wrong by more when it is wrong, carrying the largest share of gross errors at 8.7 % of plots above 50 percentage points. That is a product resolving each plot decisively where the models fitted to it hedge, which costs them where CLMS is nearly exact and saves them where it fails badly; MAE rewards the first behaviour and RMSE penalises the second, so the two metrics measure the two halves of one trade rather than contradicting each other.

Table 5: Distribution of per-plot absolute error, Milan, strict rule, $n = 450$. “Within 5 pp” and “over 50 pp” are the percentages of plots whose absolute error falls below and above those thresholds.

Map	MAE	RMSE	Median error	p90	Max	Within 5 pp (%)	Over 50 pp (%)
S2 stack	16.55	24.64	10.38	39.37	96.26	32.7	6.7
S2 percentile	16.49	24.81	9.09	40.76	94.92	34.2	6.7
emb_RF	18.30	25.63	14.38	43.67	82.84	29.8	7.3
S2 median	18.36	25.98	12.17	41.49	94.41	21.8	6.9
CLMS	14.61	26.25	3.00	45.20	100.00	53.3	8.7

What this measures is label quality. It says that the products these models were fitted to disagree with careful photo-interpretation by about as much as the models themselves do, which bears on where further gains are likely to come from. It does not measure a ceiling on achievable model performance, and nothing here should be read as one: a model is not confined to the accuracy of its labels. Section 7.3 gives the mechanism and measures it.

Figure 14 gives the plot-level view behind these summaries, every registered map in all three cities on common axes. It is the one place the scatter itself is shown rather than reduced to a metric, and the shape of the Vietnamese zero-shot panels, mass held away from the 1:1 line at low reference values, is the distributional failure Section 7.2 measures.

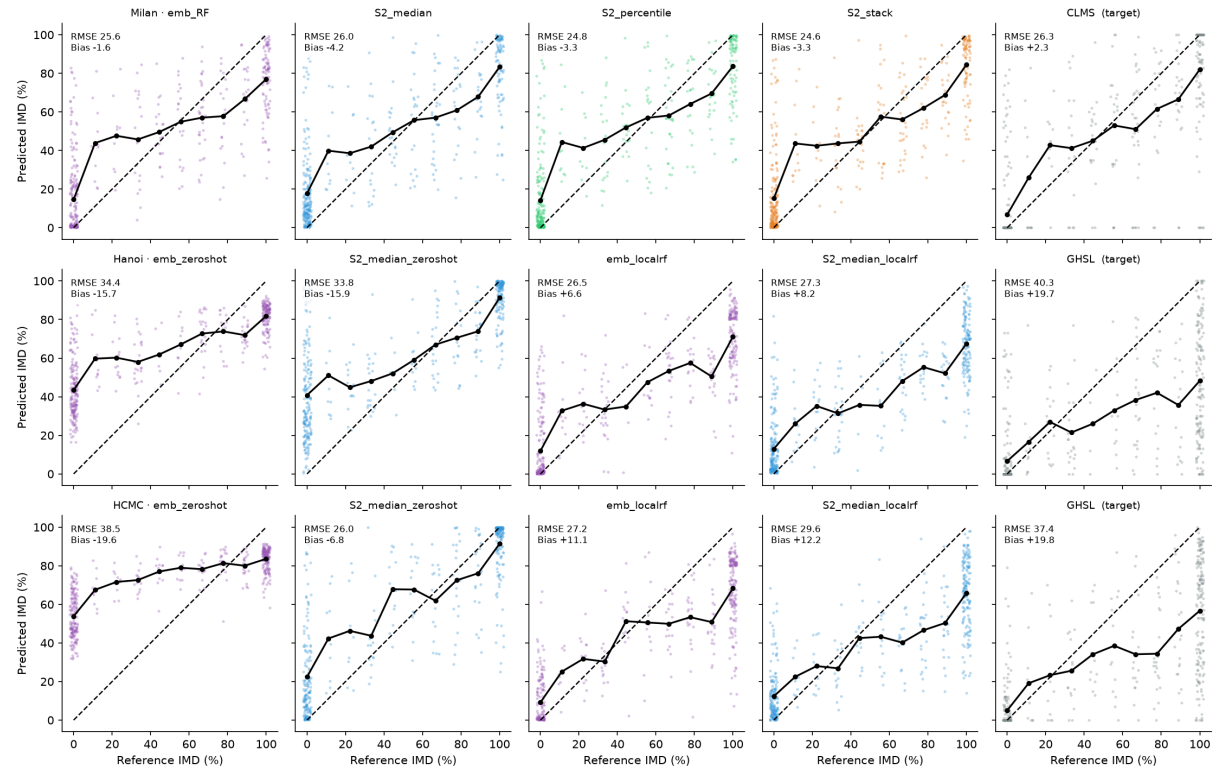


Figure 14: Photo-interpreted reference against predicted IMD, every registered map in the three cities, strict rule.

6.2 Two tiers under paired testing

The Milan four are not a clean ranking of four maps but two tiers of two: S2 stack with S2 percentile, and emb_RF with S2 median, indistinguishable within each tier and separated between them.

This analysis is independent validation and belongs here rather than in Section 4. It runs on per-plot absolute error against the photo-interpreted reference, so placing it beside the CLMS-scored holdout numbers of Section 4 would merge the two validations that Section 3.4 keeps apart.

Table 6 gives the six Milan pairs, tested by paired Wilcoxon signed-rank [14] on per-plot absolute error with Benjamini-Hochberg control of the false discovery rate [15] within the city.

Table 6: Paired Wilcoxon tests on per-plot absolute error, Milan, strict rule, $n = 450$ plots. Median difference is the median of the paired per-plot differences in absolute error.

Map A	Map B	MAE A	MAE B	Median difference	p	q (BH)	Distin-guishable
S2 median	S2 percentile	18.36	16.49	0.92	5.95e-09	3.57e-08	yes
S2 median	S2 stack	18.36	16.55	1.07	4.33e-05	1.30e-04	yes
emb_RF	S2 stack	18.30	16.55	0.60	9.62e-04	1.92e-03	yes
emb_RF	S2 percentile	18.30	16.49	0.31	4.56e-03	6.84e-03	yes
S2 stack	S2 percentile	16.55	16.49	0.19	6.55e-01	7.86e-01	no
emb_RF	S2 median	18.30	18.36	-0.64	7.90e-01	7.90e-01	no

Each of S2 stack and S2 percentile separates from each of emb_RF and S2 median, with q between 3.57e-08 and 6.84e-03. Neither of the two within-tier pairs separates: $q = 0.79$ for S2 stack against S2 percentile, and $q = 0.79$ for emb_RF against S2 median.

Two points need stating explicitly, because Section 6.1 has just shown overlapping confidence intervals for these same maps.

The first is that the paired test and the confidence intervals measure different quantities, and both are correct. An interval describes the uncertainty of one map’s RMSE taken on its own, across resamples of the plots. The paired test compares two maps on the same 450 plots, which removes the plot-level variance common to both and therefore has more power to detect a consistent difference. Overlapping intervals together with a significant paired difference is the expected signature of a small difference that is nevertheless consistent plot by plot, and not a contradiction.

The second is that the collapse reported in Section 6.1 and the tier structure reported here are both true. The four-way spread falls from 4.661 RMSE on the same-source validation, 33.0 % of the worst map at 14.123, to 1.340 on the independent validation, 5.2 % of the worst map at 25.984, a compression of about 6.4 times. The spread shrinks sharply; it does not vanish. What survives is small relative to each map’s own error and still resolvable across 450 paired plots. Neither half of that statement may be dropped.

6.3 Rule sensitivity

The conclusion does not depend on how the impervious classes are coded. Table 7 gives the Milan RMSE under all three rules with the resulting ranks.

Under rule B the ranking is unchanged. Rule B reclassifies 23 cells across 6 plots in Milan, and rule C reclassifies 112 cells across 18 plots, so B is much the smaller perturbation. In Hanoi and HCMC the permeable-pavement code does not occur at all, so rule B is identical to strict there by construction and only rule C differs.

Under rule C the top two maps and CLMS hold their positions, while emb_RF and S2 median exchange

Table 7: Milan RMSE against photo-interpretation under the three impervious rules. Rule B adds permeable pavement to the strict coding, rule C adds unpaved dirt road. Paved roads are impervious under all three.

Map	Strict	B	C	Rank strict	Rank B	Rank C
S2 stack	24.64	24.54	20.98	1	1	1
S2 percentile	24.81	24.71	21.37	2	2	2
emb_RF	25.63	25.54	23.03	3	3	4
S2 median	25.98	25.94	22.98	4	4	3
CLMS	26.25	25.97	24.08	5	5	5

third and fourth: 23.03 against 22.98, a difference of 0.05. The two maps that swap are exactly the pair that Section 6.2 finds statistically indistinguishable at $q = 0.79$. The swap is therefore a reordering within noise and not a change of result.

All five Milan maps improve under rule C, by between 2.18 and 3.67 RMSE, and CLMS is among them. Counting compacted dirt roads as impervious moves every map closer to the reference, including the training target itself, which indicates that the disagreement rule C resolves is in the reference coding rather than in any particular map.

7 Discussion

7.1 From same-source to independent validation

Most of the separation between predictor sets that the same-source validation reports is agreement with CLMS rather than accuracy. A real and resolvable difference nonetheless survives the change of reference.

The size of the effect is the clearest way to see it. On the same-source validation the percentile composite leads the AlphaEarth embeddings by 4.661 RMSE, 9.462 against 14.123. On the independent validation the same two maps are 24.809 and 25.625, a lead of 0.816. Across all four predictor sets the spread falls from 4.661, which is 33.0 % of the worst map at 14.123, to 1.340, which is 5.2 % of the worst map at 25.984. Relative to each validation’s own error scale, that is a compression of about 6.4 times. [Figure 15](#) shows the four maps moving between the two validations.

Part of the collapse is a reshuffle rather than a compression. The worst map differs between the two validations: the embeddings are last on the same-source validation at 14.123, while on the independent validation the S2 median is last at 25.984 and the embeddings sit third at 25.625. An ordering that survived three metrics within the first validation does not survive the change of reference intact.

The collapse should not be overstated in the other direction either. Section 6.2’s paired testing resolves two tiers on exactly these data, with q between $3.57\text{e-}08$ and $6.84\text{e-}03$ between tiers and $q = 0.79$ within each. The residual difference is therefore real and detectable, not lost in noise. Both halves of that belong in the same sentence: the advantage of the better composites shrinks sharply but does not disappear. It is small relative to each map’s own error and consistent enough across 450 paired plots to be significant.

The methodological conclusion follows from that pairing. The same-source validation remains the right instrument for model selection: it is available for every candidate, it uses the full spatial holdout, and it ranks the four predictor sets in an order that the independent validation broadly, if not exactly, endorses at the top. It is the wrong instrument for claiming accuracy, because most of what it measures as separation is the degree to which each map has learned to resemble CLMS. A 33 % reduction in same-source RMSE is a statement about agreement with a particular product, and reporting it as a 33 % gain in accuracy would misdescribe it by a factor of six.

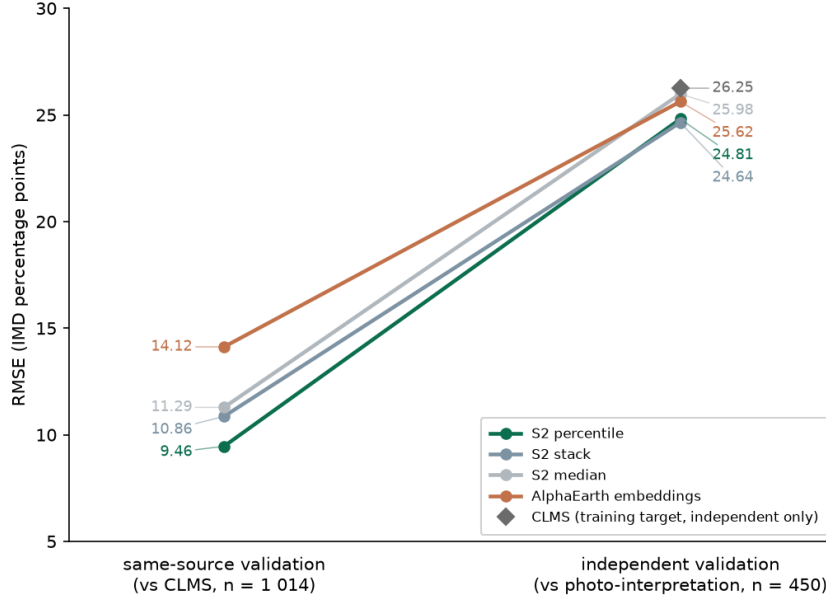


Figure 15: The four Milan predictor sets, same-source RMSE against CLMS on the 1 014-point holdout on the left, independent RMSE against the 450 photo-interpreted plots on the right. CLMS is marked on the independent axis as a map under test. The two axes are different measurements against different references and the connecting lines trace each map’s position within each of them, not a change in its error.

7.2 Why transfer fails: level matching and the lost low tail

The transferred maps fail because their prediction level does not match the target city, and because zero-shot transfer destroys the low end of the predicted distribution. The second mechanism is the more damaging of the two and holds in both cities.

The level argument is visible in the mean predicted values against the interpreted reference. Both zero-shot maps inherit Milan’s high level: 61.77 and 61.93 in Hanoi against a reference mean of 46.05, and 70.70 and 57.89 in HCMC against a reference mean of 51.06. Both local retrains inherit GHS-BUILT-S’s low level, predicting between 37.87 and 39.97 across the two cities. The two scenarios therefore straddle the reference from opposite sides, and which of them scores better in a given city depends partly on where that city’s own reference level happens to fall between them.

Ho Chi Minh City is the case where this matters. The S2 median zero-shot map records the city’s best independent RMSE at 25.950, ahead of both local retrains at 27.242 and 29.615. Its level gap is 6.83, the smallest of the four HCMC maps, where the local retrains sit 11.09 and 12.23 below the reference. The zero-shot map wins there because Milan’s level happens to land closer to HCMC’s reference level than GHSL’s does, not because the transfer worked. In Hanoi, where the reference mean is lower at 46.05 and the zero-shot maps overshoot by close to 16 points, the same two scenarios reverse and the local retrains lead by six to eight RMSE. A ranking that flips with the target city’s level is a ranking set by level matching.

The stronger evidence is the shape of the predicted distribution, because it does not depend on level at all. Two properties are measured separately here: how much of the reference’s low tail a map retains, and how much of its spread. The first is the share of the reference’s sub-20 % mass the predicted raster reproduces, the second the map’s interquartile range against the reference’s. They are reported as measured quantities rather than as a threshold verdict, because the two do not have to move together and here they do not.

Both embeddings zero-shot maps have lost the low tail. Hanoi retains 3.1 % of the sub-20 % mass its reference carries and HCMC 0.0 %; below 10 % both rasters are empty to four decimal places. This is not a displaced distribution in either city, because a shifted map would keep its spread and move only its

Table 8: Low-tail and spread retention for the eight Vietnamese maps, computed over the full predicted rasters, 8.4 million valid pixels in Hanoi and 8.2 million in HCMC. Tail retained is the map’s share of pixels below 20 % IMD as a proportion of the reference’s share of plots below 20 %; spread retained is the map’s interquartile range as a proportion of the reference’s. The tail figure compares a raster-wide share against a 450-plot share, so it is read as an order of magnitude rather than to the percentage point, and a value at or above 100 % means the tail is fully present rather than oversized.

City	Map	Scenario	Tail retained (%)	Spread retained (%)
Hanoi	emb_zeroshot	zero-shot	3.1	37
Hanoi	S2_median_zeroshot	zero-shot	30.6	53
Hanoi	emb_localrf	local retrain	136.0	61
Hanoi	S2_median_localrf	local retrain	124.4	51
HCMC	emb_zeroshot	zero-shot	0.0	26
HCMC	S2_median_zeroshot	zero-shot	68.7	75
HCMC	emb_localrf	local retrain	108.3	57
HCMC	S2_median_localrf	local retrain	104.0	51

centre. The loss is therefore a property of the transferred embedding map rather than of one city, which is what makes it a mechanism rather than an anecdote.

The two cities differ in how far the spread collapses alongside the tail. HCMC’s map holds 26 % of the reference interquartile range against Hanoi’s 37 %, on IQRs of 26.0 and 37.4 against a reference IQR of 100.0, and its standard deviation is 15.95 against the reference’s 44.81. HCMC is the severe case: the range has narrowed around the missing tail as well as losing it. That severity is visible in the scores, where the HCMC embeddings zero-shot map is the worst of all fifteen at 38.494 despite a mean of 70.70 that is not absurd for the city.

The consequence is that no correction of level could repair either map. Both references are strongly bimodal, 41.6 % of Hanoi plots below 10 % and 31.8 % above 90 %, 37.3 % and 35.8 % in HCMC, and a map with no low tail cannot represent the lower mode wherever its centre is placed. Shifting such a map downward would move its mass off the upper mode without populating the lower one.

The complement of that result is what the local retrains do. All four reach a raster floor of 0.00 and retain between 104.0 % and 136.0 % of the reference’s sub-20 % mass, which is full recovery of the low end within the precision this comparison supports. The two mechanisms then state the failure cleanly and symmetrically: zero-shot transfer destroys the low end of the distribution and local retraining restores it, in both cities and for both predictor sets. That is a stronger statement than either city alone would support, and it is the distributional counterpart of the R^2 recovery reported in Section 5.1.

The S2 median zero-shot maps sit between the two, retaining 30.6 % of the tail in Hanoi and 68.7 % in HCMC. They are degraded rather than emptied, which is consistent with Section 5.1’s finding that both predictor sets fail under transfer while the embeddings fail harder. Section 8’s caveat applies to that contrast: the two zero-shot experiments differ in composite depth as well as in predictor type, so the ordering between them is not a controlled comparison.

Figure 16 carries the argument for HCMC: the predicted distributions for the four maps against the reference, with the level means annotated.

One comparison this section does not make is between the two Hanoi S2 median scenarios. The paired test of S2_median_zeroshot against S2_median_localrf in Hanoi reaches $q = 0.063$ and is recorded as a non-result: at 450 plots the test does not separate them. That is not evidence that the two are equivalent. Absence of a detected difference is not a demonstration of no difference, and nothing in the transfer argument rests on that pair.

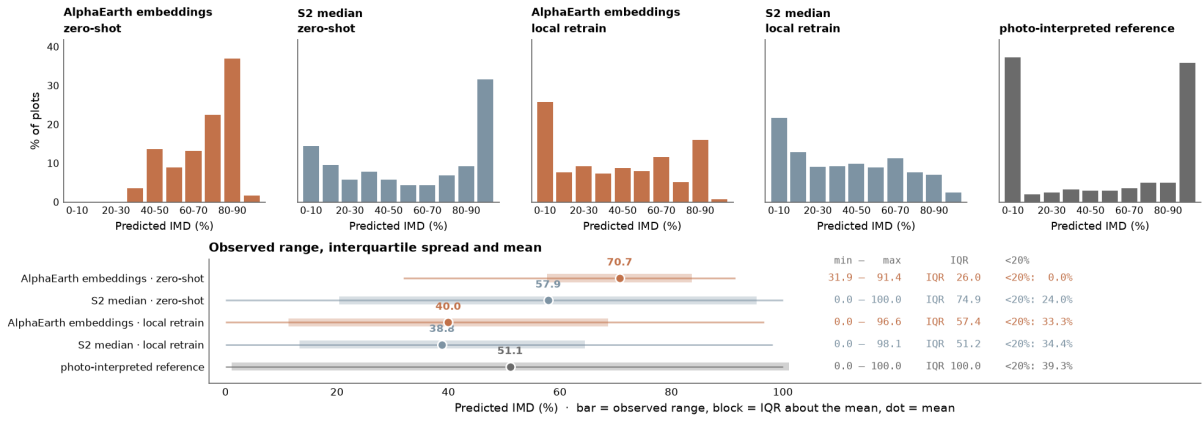


Figure 16: Distribution of predicted IMD over the 450 HCMC plots for the four maps and the photo-interpreted reference, strict rule, with each map’s mean level marked. The reference is bimodal; the embeddings zero-shot distribution is confined to a narrow band with no low tail. HCMC is shown because it is the severe case, but Table 8 records that the tail is lost in Hanoi as well.

7.3 Retraining works: bias recovery

Local retraining does not merely reproduce its training target’s offset. It recovers between a third and two thirds of it, which means the fitted models extract signal their labels do not carry.

The effect itself is not new here. Žgela reports it qualitatively [1, p. 10], observing that roads and unroofed impervious surfaces “are visibly better represented in the predicted maps, although still imperfectly, as the model itself was trained against the GHS-BUILT-S reference”, and that this “highlights the ability of the model to recover additional information beyond what was explicitly provided during training”. He established it by visual inspection of his Fig. 11. The contribution of this project is to measure it against an independent reference, which his study had no means to do.

Table 9 gives the measurement.

Table 9: Bias against photo-interpretation for GHS-BUILT-S and for the local retrains fitted to it, with the share of the target’s deficit that each map recovers. Bias is observed minus predicted, so a positive value means the map under-marks the interpreted reference.

City	Target	Target bias	Map	Map bias	Recovered (pp)	Recovered (%)
Hanoi	GHSL	19.68	emb_localrf	6.59	13.09	66.5
Hanoi	GHSL	19.68	S2_median_localrf	8.18	11.50	58.4
HCMC	GHSL	19.80	emb_localrf	11.09	8.71	44.0
HCMC	GHSL	19.80	S2_median_localrf	12.23	7.57	38.2

GHS-BUILT-S under-marks by 19.68 percentage points in Hanoi and 19.80 in HCMC. The models fitted to it are biased by only 6.59 to 12.23, recovering 7.57 to 13.09 percentage points, or 38 % to 67 % of the target’s systematic deficit. The recovery is larger in Hanoi than in HCMC and larger for the embeddings than for the S2 median in both cities, but it is substantial in all four cases. Figure 17 shows the four movements.

This is the mechanism behind the training-product result of Section 6.1, and it is what keeps that result from meaning something it does not. Section 6.1 shows that the training products score no better than the models fitted to them, and states that this measures label quality rather than a ceiling on achievable performance. The bias recovery is why the two statements are consistent. A model fitted to a noisy target does not reproduce the target. Label error that is random from pixel to pixel cannot be fitted by

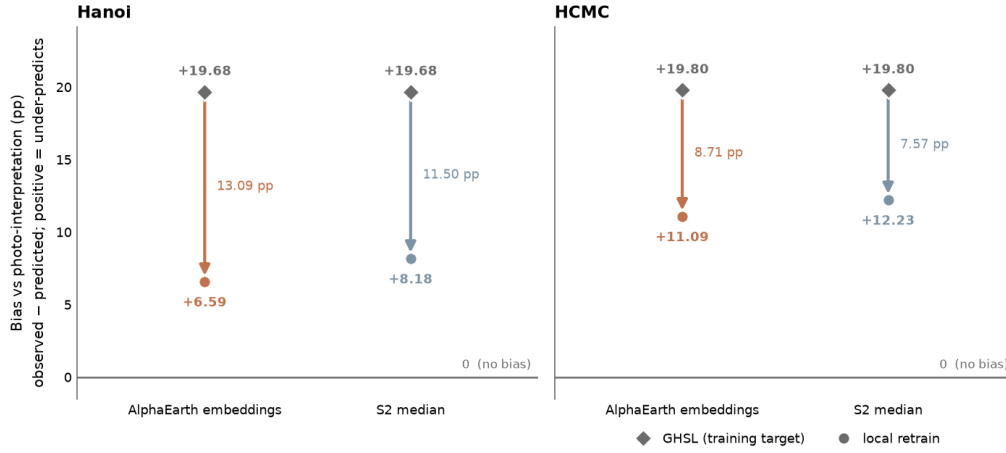


Figure 17: Bias against photo-interpretation, from GHS-BUILT-S to each local retrain fitted to it, Hanoi and HCMC. Each arrow spans the share of the reference’s systematic deficit that the model recovers.

any model with limited capacity, so it is smoothed away rather than learned, and the fitted surface lands closer to the truth than the individual labels do. Systematic label error is different, and this is where the second mechanism enters: GHS-BUILT-S’s deficit is systematic, arising from its exclusion of roads by design (Section 2.1), yet the predictors still respond to the sealed surfaces the target omits. A road that GHS-BUILT-S does not mark is nonetheless visible to the embeddings and to the Sentinel-2 reflectance percentiles, and a model trained on plots where roads happen to co-occur with marked built-up area learns a response that partly recovers them.

Neither mechanism recovers the deficit fully, and the numbers say so: 38 % to 67 %, not 100 %. The residual bias of +6.59 to +12.23 is the part of the target’s systematic error the models do inherit. The claim is bounded accordingly. A model can score better than its own training target, and here it measurably does, but it is not thereby freed from that target: retraining on a product that under-marks by 20 points still yields maps that under-mark by 7 to 12.

8 Limitations

Five constraints bound what the results above support. Each is stated with the conclusion it forbids, because a limitation that does not change a reading of the results is not worth recording.

Composite depth is confounded with predictor type in Vietnam. The Milan median composite draws on the thirty usable dates that survive cloud screening, while Hanoi has four and Ho Chi Minh City three, of which two are five days apart. The AlphaEarth zero-shot experiment therefore compares an annual embedding in Milan against an annual embedding in Vietnam, whereas the Sentinel-2 zero-shot compares a thirty-date composite against a three or four-date one. The two zero-shot arms are not equally handicapped. What this forbids is the comparison between them: nothing in Section 5.1 or Section 7.2 supports a statement that the embeddings transfer worse than Sentinel-2, or better, because the difference between the two arms in Vietnam is inseparable from the difference in composite depth. The shared conclusion, that both fail and that local retraining repairs both, is unaffected, because it does not rest on the ordering between them. Section 2.3 gives the surviving dates per city.

Plot-level independence is assumed and untested. The 450 plots in each city enter every metric, every bootstrap interval and every paired test as independent observations. No spatial autocorrelation check was applied at plot level, and the sampling design does not guarantee independence: plots drawn from the same neighbourhood share both land cover and, plausibly, correlated model error. If residual autocorrelation is present, the effective sample size is below 450, the bootstrap intervals in Fig. 13 are narrower

than they should be, and the paired q values of Section 6.2 are optimistic. This bears most directly on the two-tier result. The tier separation runs from $q = 3.57\text{e-}08$ to $q = 6.84\text{e-}03$, and the stronger of those would survive a considerable inflation of the standard errors, but the weakest would not necessarily survive a large one. The safe reading is that the tier structure is supported and that its weakest pair should not be treated as established at the quoted precision.

RMSE_{corr} is an upper bound on map error, not a point estimate. The correction subtracts a binomial estimate of reference noise built on the assumption that the nine sub-cells of a plot are independent draws. They are not: sub-cells within a single 10 m pixel share land cover and are spatially correlated, so the binomial variance overstates nothing but understates the true reference noise, and the correction consequently removes less error than it should. RMSE_{corr} therefore bounds map error from above and cannot be quoted as the error of any particular map. Section 6.1’s statement that the four Milan maps lie between 21.559 and 22.917 corrected is a statement about bounds, and the claim that any of them achieves an error of 21.559 is not supported.

A single interpreter produced the reference, and there is no inter-rater agreement estimate. All 1 350 plots were interpreted by one analyst under one reading of the coding rules. Systematic interpretation tendencies, such as where the boundary between built and unbuilt is drawn in dense mixed fabric or how partially shadowed sub-cells are resolved, are therefore common to the whole reference rather than averaged out across interpreters, and no measurement of their size exists. The rule sensitivity analysis of Section 6.3 addresses only the coding choice that was varied deliberately, not interpreter variation within a fixed coding.

The scope of the resulting caveat is narrow and should not be widened. Because the paired tests of Section 6.2 difference two maps plot by plot against the same interpreted value, an interpretation tendency that displaces a plot’s reference value affects both maps in the pair and drops out of that difference, and the tier structure is robust to it to that extent. No such cancellation applies anywhere else. The absolute RMSE and MAE levels of Table 4 carry an unquantified interpreter component in full, as do the bootstrap intervals of Fig. 13 and the bias measurements of Section 7.3, because each of those scores a map against the reference rather than against another map. Nor does the shared reference protect the rankings: a tendency that displaces the reference by different amounts in different kinds of plot penalises maps unequally, so the ordering in Table 4 is not immune either. The single claim the shared reference supports is that the paired within-city differences are less exposed to interpreter bias than the error levels are.

The plots are stratified by the product under test, so the metrics are not city-wide accuracy. Plot locations were drawn under a stratification over the reference product rather than at random over the city, and no area weights are applied when the per-plot errors are aggregated. Every number in Section 6 is therefore an unweighted mean over a stratified sample, not an estimate of the error a user would encounter over the city as a whole. Strata that occupy a small share of the urban area are represented as heavily as strata that occupy a large one, and the bimodality of the Vietnamese references reported in Section 7.2 is a property of the plot sample and not necessarily of the cities. The comparisons between maps hold, because every map is scored on the same plots. The statement that would be wrong is that any of these RMSE values describes the accuracy of a map over its city.

9 Conclusions

Explicit Sentinel-2 composites outperform the AlphaEarth embeddings against CLMS in Milan by a wide margin. Against independent photo-interpretation that margin compresses sharply but remains resolvable. Zero-shot transfer to Hanoi and Ho Chi Minh City fails, and local retraining is required. Both training products score no better than the models fitted to them, which measures the quality of the labels and not a limit on what the maps can achieve.

1. Composite choice matters more than the foundation model, on the same-source validation. The

percentile composite reaches an RMSE of 9.462 against the embeddings' 14.123, a reduction of 33 %, and the ordering of the four predictor sets is the same on RMSE, MAE and R^2 . The advantage is mechanistically interpretable rather than merely empirical: B4_p25 alone carries a permutation importance of 0.1812, more than three times that of the next band, and four of the top five bands are low percentiles of red or high percentiles of near infrared, precisely the quantiles a median composite discards (Section 4.3). The embeddings offer no comparable account of why they work.

2. Against an independent reference the four-way spread compresses about 6.4-fold, from 33.0 % to 5.2 % of the worst map, yet paired testing still resolves two tiers. Both halves of that sentence are required. The spread falls from 4.661 RMSE on the same-source validation to 1.340 on the independent one, so most of what the same-source validation measured as separation between predictor sets was agreement with CLMS rather than accuracy. What survives is nonetheless real: S2 stack and S2 percentile each separate from emb_RF and S2 median at q between $3.57\text{e-}08$ and $6.84\text{e-}03$, while neither within-tier pair separates at $q = 0.79$ (Section 6.2). The ordering is also partly reshuffled, not merely compressed. The S2 median is last on the independent validation at 25.984 where the embeddings were last on the same-source one at 14.123. The same-source validation remains the right instrument for model selection and the wrong one for claiming accuracy.

3. Zero-shot transfer fails and local retraining is necessary. The zero-shot R^2 is at or below zero in three of the four city and predictor combinations, -0.070 , -0.142 and -0.279 , recovering to between 0.522 and 0.649 when the model is retrained locally. The failure has two separable mechanisms. The transferred maps inherit Milan's level, predicting means of 61.77 and 61.93 in Hanoi and 70.70 and 57.89 in HCMC against reference means of 46.05 and 51.06, so which scenario scores better in a city is set partly by where that city's reference level falls. The more damaging mechanism is distributional and holds in both cities: the embeddings zero-shot maps retain 3.1 % of the reference's sub-20 % mass in Hanoi and 0.0 % in HCMC, and every local retrain restores it, retaining between 104.0 % and 136.0 % (Section 7.2). Retraining is not uniformly better, however. Above 80 % imperviousness it is worse: the embeddings model's C6 MAE rises from 17.25 to 30.95 in Hanoi and from 12.93 to 25.20 in HCMC (Section 5.2).

4. The training products are not ceilings. CLMS scores an RMSE of 26.253 against photo-interpretation and GHS-BUILT-S 40.345 in Hanoi and 37.355 in HCMC, in each case no better than the models fitted to them. What that measures is label quality: it indicates that further gains are unlikely to come from features or estimators alone while the labels disagree with careful interpretation by this much. It does not bound achievable accuracy, and the measurement that shows why is in Section 7.3. GHS-BUILT-S under-marks the interpreted reference by 19.68 percentage points in Hanoi and 19.80 in HCMC, while the local retrains fitted to it are biased by only 6.59 to 12.23, recovering 7.57 to 13.09 points, or 38 % to 67 % of the deficit. A model fitted to these labels measurably outperforms them. The recovery is bounded rather than complete, and the bound is part of the conclusion: retraining on a product that under-marks by roughly 20 points still yields maps that under-mark by 7 to 12.

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